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Sec: B

```
%{
#include "y.tab.h"
#include <string.h>
%}

%%
[0-9]+              { yylval.str = strdup(yytext); return NUM; }
[a-zA-Z][a-zA-Z0-9]* { yylval.str = strdup(yytext); return ID; }
[ \t\n]+            ;
[+\-*/()]           { return yytext[0]; }
.                   { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```

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```
      }
    | expr '*' expr {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = %s * %s\n", $$, $1, $3);
      }
    | expr '/' expr {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = %s / %s\n", $$, $1, $3);
      }
    | '(' expr ')' {
          $$ = $2;
      }
    | '-' expr %prec UMINUS {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = -%s\n", $$, $2);
      }
    | ID  { $$ = $1; }
    | NUM { $$ = $1; }
    ;

%%

int main()
{
    printf("Enter arithmetic expression:\n");
    yyparse();
    return 0;
}

int yyerror(char *s)
{
    printf("Invalid expression\n");
    return 0;
}
```

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```
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int yylex(void);
int yyerror(char *s);
int temp = 1;
%}

%union {
    char *str;
}

%token <str> ID NUM
%type <str> expr

%left '+' '-'
%left '*' '/'
%right UMINUS

%%

input:
    expr {
        printf("Final result = %s\n", $1);
    }
    ;

expr:
      expr '+' expr {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = %s + %s\n", $$, $1, $3);
      }
    | expr '-' expr {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = %s - %s\n", $$, $1, $3);
      }
    | expr '*' expr {
          $$ = malloc(20);
          sprintf($$, "t%d", temp++);
          printf("%s = %s * %s\n", $$, $1, $3);
      }
```

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```
div12@div12:~/CC lab$ yacc -d Exp12.y
div12@div12:~/CC lab$ flex Exp12.l
div12@div12:~/CC lab$ gcc y.tab.c lex.yy.c -o Exp12 -ll
div12@div12:~/CC lab$ ./Exp12
Enter arithmetic expression:
a+b*c
t1 = b * c

t2 = a + t1
Final result = t2
```

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```
%{
#include "y.tab.h"
%}

%%
"int"                   { return INT; }
[a-zA-Z][a-zA-Z0-9]*    { return ID; }
","                     { return COMMA; }
";"                     { return SEMICOLON; }
[ \t\n]+                ;
.                       { return yytext[0]; }
%%

int yywrap(void)
{
    return 1;
}
```

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```
div12@div12:~/CC lab$ yacc -d Exp11.y
div12@div12:~/CC lab$ flex Exp11.l
div12@div12:~/CC lab$ gcc y.tab.c lex.yy.c -o Exp11 -ll -lm
div12@div12:~/CC lab$ ./Exp11
Enter a variable declaration:
int a,b,c;
Valid declaration
```

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```
%{
#include <stdio.h>
#include <stdlib.h>
-
int yylex();
void yyerror(const char *s);
%}

%token INT ID COMMA SEMICOLON

%%
declaration:
    INT identifier_list SEMICOLON
    {
        printf("Valid declaration\n");
    }
    ;

identifier_list:
    ID
    | identifier_list COMMA ID
    ;
%%

void yyerror(const char *s)
{
    printf("Invalid declaration\n");
}

int main()
{
    printf("Enter a variable declaration:\n");
    yyparse();
    return 0;
}
```